

Basic Machine Learning – II

Session Agenda

Duration: 3 Hours

Session Overview

This session introduces several fundamental machine learning concepts that collectively enable intelligent AI systems. Participants begin by understanding Decision Trees and Random Forests for prediction and classification tasks, then explore Clustering for discovering hidden patterns in unlabeled data — including how to evaluate clustering quality without labels — and Feature Engineering for improving model learning. The session closes with Overfitting and Underfitting, and Optimization Basics and Loss Functions — the mathematical foundation that allows machine learning models to learn from data and continuously improve their predictions, connecting these ideas to modern AI and Agentic AI applications.

Learning Outcomes

- Explain Decision Trees and Random Forests, and how ensembles improve on a single tree
- Differentiate supervised and unsupervised learning
- Apply clustering for customer/data segmentation problems
- Evaluate clustering quality using the Silhouette Score and, when ground truth exists, the Adjusted Rand Index
- Perform basic feature engineering to improve model learning
- Identify overfitting and underfitting scenarios using the bias-variance tradeoff
- Understand optimization and gradient descent concepts
- Explain the role of loss functions in machine learning

Detailed Session Schedule

Duration	Topic / Content
Part 1 — Tree-Based Models & Clustering (85 min)	
10 min	Recap of Session 1 + Agenda: Quick recap of regression, classification, SVM, and evaluation metrics; agenda for today.
20 min	Decision Trees: Flowchart-of-questions analogy, splitting criteria (Gini/Entropy at intuition level), pros and cons.
15 min	Random Forest: Ensemble/voting idea — combining many trees, and why it reduces overfitting compared to a single tree.
25 min	Practical 1: Decision Tree and Random Forest classification demo, with a feature importance plot.
15 min	Clustering: K-Means intuition, unsupervised vs supervised learning, and customer segmentation as a use case.
10 min	<i>Break</i>
Part 2 — Clustering Practical, Feature Engineering & Optimization (85 min)	
10 min	Model Evaluation Metrics: Clustering: Evaluating cluster quality without labels using the Silhouette Score, and with ground truth using the Adjusted Rand Index.

15 min	Practical 2: K-Means clustering on a customer segmentation dataset, with cluster visualization and evaluation using the Silhouette Score.
15 min	Feature Engineering: Creating, transforming, and selecting features; binning and interaction features.
15 min	Practical 3: Hands-on feature engineering exercise applied to a beginner-friendly dataset.
10 min	Overfitting & Underfitting: Bias-variance tradeoff, how to detect each scenario, and a brief introduction to regularization.
10 min	Optimization Basics & Loss Functions: Gradient descent intuition, MSE vs cross-entropy loss, and how these connect to Agentic AI/GenAI pipelines.
10 min	Recap + Q&A + Wrap-up: Key takeaways across both sessions and open discussion.